Data, Information, Knowledge, Wisdom — DIKW

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DIKW and MLDM as-a-process

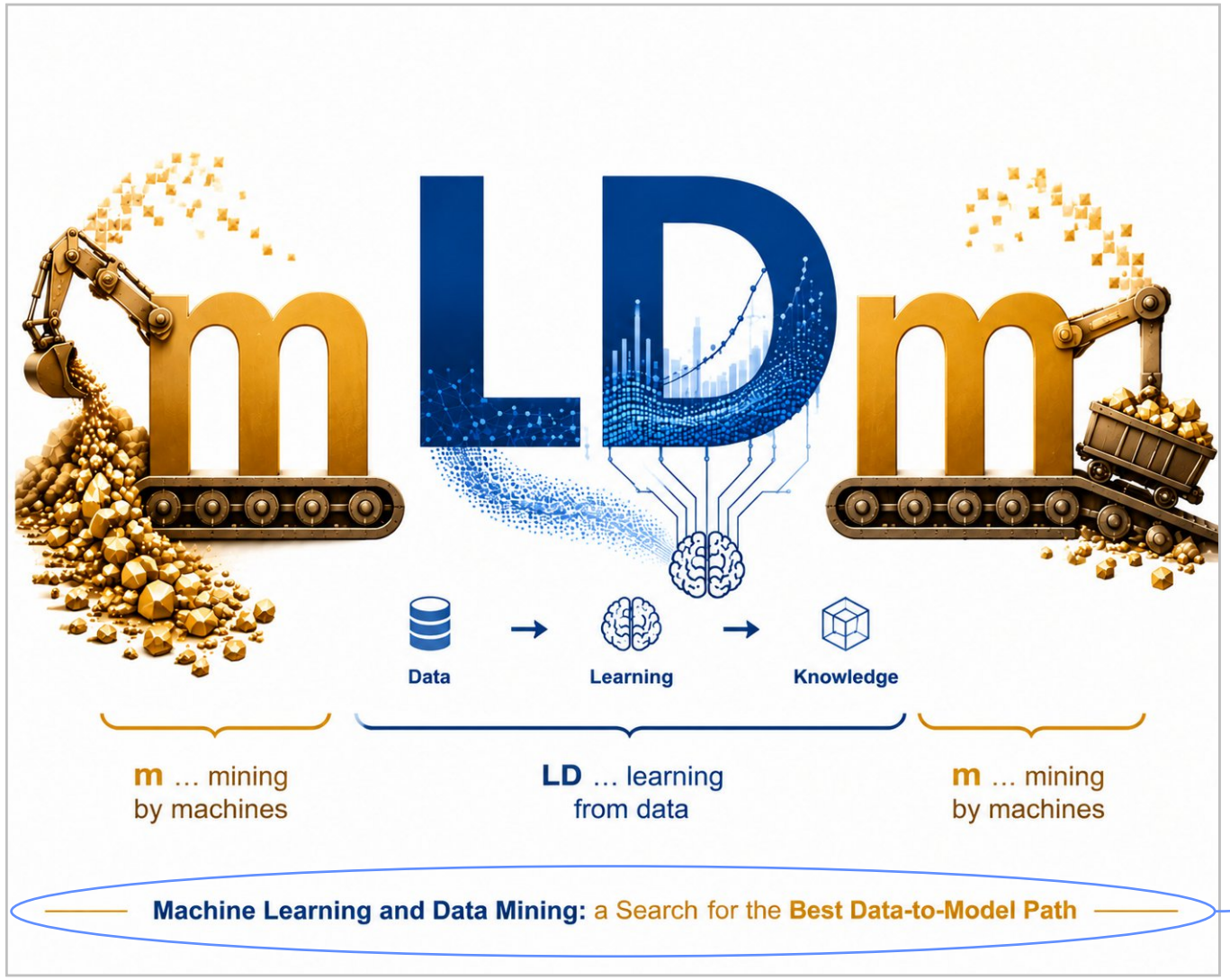


image created by a generative AI (gAI) model

machine-Learning and Data-mining

mLDm

as-a-process **that searches for the**

best Data-to-Model path

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... questions and a goal

(some) questions about data, information, knowledge and wisdom:

- why these concepts matter in MLDM?
- how to relate DIKW to the stages of MLDM as-a-process?

before searching for the best data-to-model path

the goal is to grasp a conceptual view of

how *raw-observations* become *actionable-understanding*

i.e., how to go from *data* into *decision-making*

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Why begin with DIKW?

- MLDM is **not only** about algorithms
- it is a **process** involving observations, context, explanation and judgment
- DIKW **helps clarify** what is **transformed** (into what) at **each stage** of the process
- it also helps to explain why **raw data alone does not create value**

Data
raw **observations** or records

Information
data considered within a **context**

Knowledge
information **explained** by patterns

Wisdom
knowledge **judged** through action

important caution **— DIKW is a conceptual aid — it is not a rigid scientific law or a mandatory taxonomy**

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Data — the starting point

definition

raw observed or recorded values, events, measurements or other **symbols without sufficient interpretation** by themselves

examples

sensor values, transaction logs
pixels, text-strings, timestamps

key-limitation

data alone **does not explain** meaning, relevance or support decision-making

data has a **type** — numeric, categorical, text, image, temporal, spatial

and may be acquired via batch or stream

the **same data** can support **different problems** depending on context and goals

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Information — data interpreted in context

definition

data organised, cleaned, contextualised or summarised so that it answers basic questions such as **who**, **what**, **where** or **when**

examples

average by group, frequency table

joined dataset, clean timestamp meaning

transition

information arises when data is **structured and interpreted** relative to a problem

data-preparation is often the stage where raw data starts becoming useful information

in MLDM, information is already shaped by **choices about representation** and context

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Knowledge — patterns, models and justified understanding

definition

structured understanding derived from information through analysis, modelling or discovery of regularities

examples

classification rules, regression relations, clusters, association rules

transition

knowledge emerges when **patterns are extracted**, evaluated and justified

a **model is only useful knowledge** if its assumptions and evidence are defensible

in MLDM, much of the **analytical core** aims to transform information into knowledge

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Wisdom — using knowledge “well” (wisely)

definition

evaluation, or judgement, about how knowledge should be applied for decision, intervention or action

examples

formulate a strategy, intervene on a process, accept or reject a decision

important-note

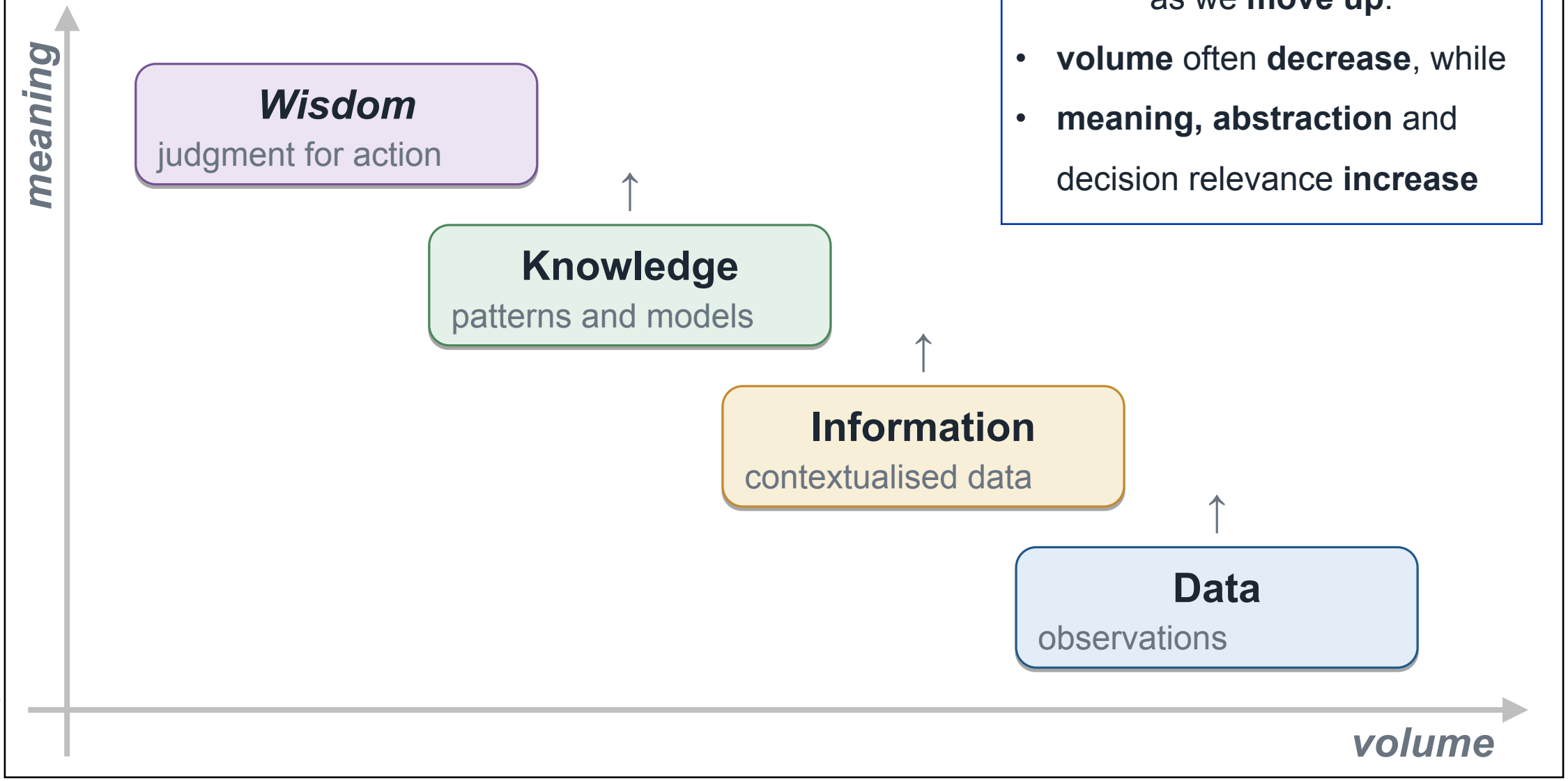
wisdom is **not produced by data mining**; it depends on human goals, constraints and values

a wise decision is **not automatically implied** by a high-performing model

MLDM may supports wisdom, but does not replace human judgment

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The DIKW ladder — a compact view



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... “climbing the ladder” is (an iterative) process

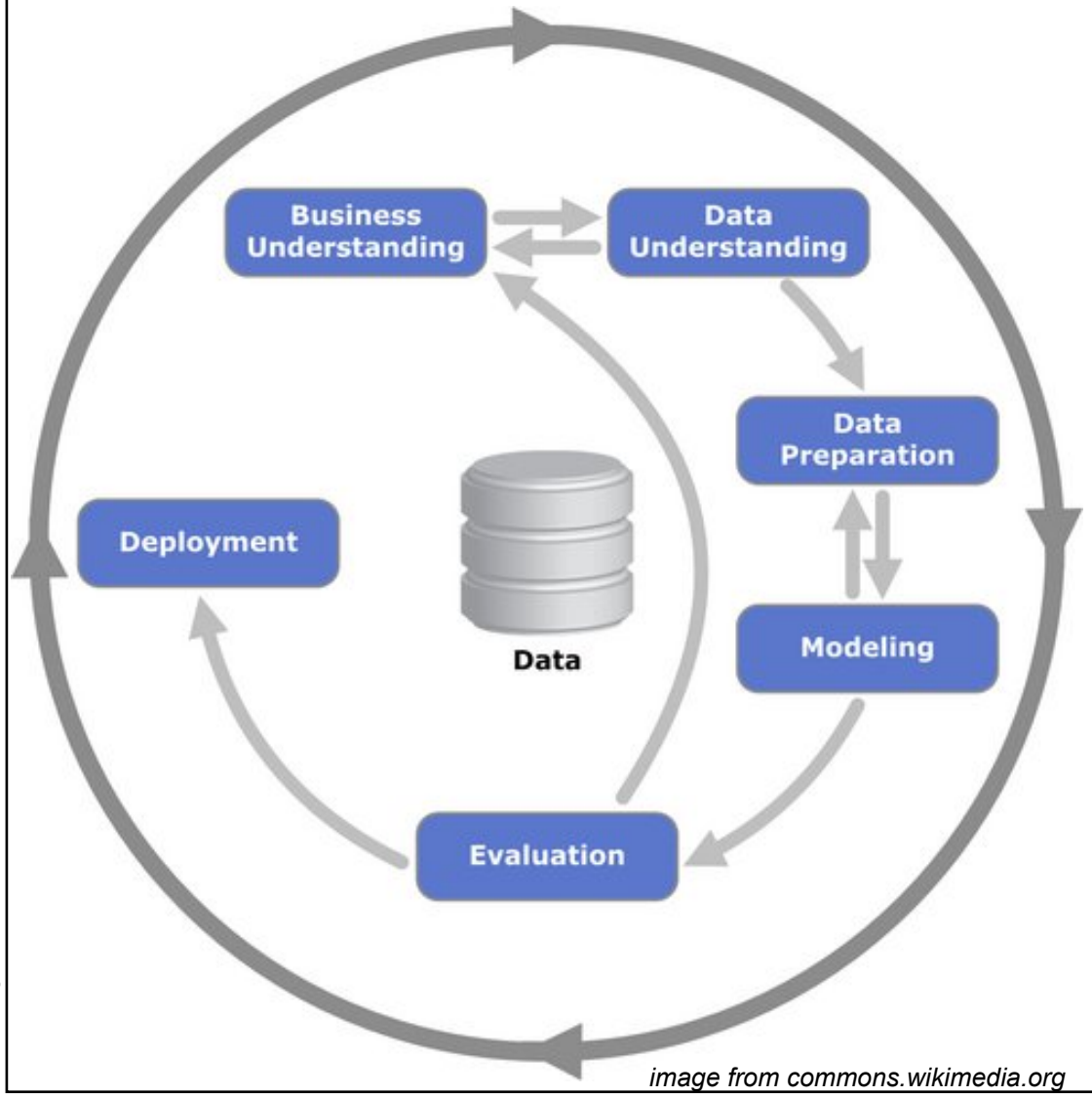


image from commons.wikimedia.org

CRISP-DM

Cross-Industry Standard Process for Data Mining

is “the classical” process

- it is **not the only possible process** design but it is a clear classical baseline
- it remains a **useful reference** because it emphasizes phases (steps) and iterations

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CRISP-DM — a classical process for data-mining

	phase	usual-task
1	*Business-Understanding*	*define goals, success criteria and constraints*
2	Data-Understanding	**collect, describe and explore** available data
3	Data-Preparation	**clean, integrate, transform** and construct **datasets**
4	Modeling	**select** methods, **train** and **test** models, tune parameters
5	Evaluation	**assess results** relative to goals and validity
6	Deployment	deliver or **operationalise** results

*in MLDM the business-understanding is relevant just from the overall motivation perspective; helps to identity success criteria and constraints; but **not from a "purely" business** perspective*

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... how CRISP-DM relates to DIKW? a conceptual articulation

CRISP-DM *phase*	**DIKW *emphasis***	***articulation-perspective***
Business-Understanding	*wisdom*	decision context, objectives, constraints
Data-Understanding	data | information	describe what exists and what it may mean
Data-Preparation	information	make data usable and structured for analysis
Modeling	information | knowledge	extract patterns and predictive models
Evaluation	knowledge | *wisdom*	judge if knowledge is valid and useful
Deployment	*wisdom-in-action*	apply results in an operational setting

CRISP-DM describes a **process**; **DIKW** describes the **nature of what is being transformed**
together, they help explain both the **workflow and the meaning** of the workflow

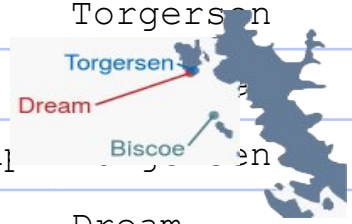
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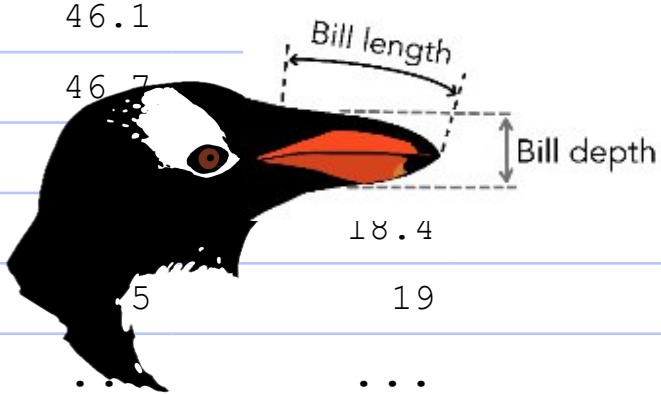
A simple illustrative example — using the “penguin” dataset

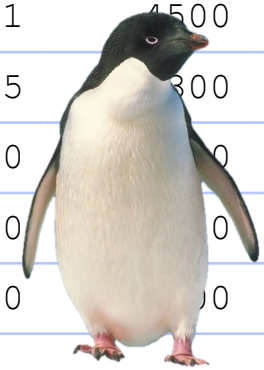
data was collected as part of research to study the Palmer Archipelago (Antarctica)
***penguins**' foraging behavior and its relationship with environmental variability*

species	island	bill length_mm	bill depth_mm	flipper length_mm	body mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie		<null>	17.9	190	3325	female	2009
Chinstrap		52	19	197	4150	male	2007
Adelie	Dream	36.8	18.5	<null>	3500	female	<null>
Adelie	Dream	41.5	18.5	201	4000	male	2009
Gentoo		46.1		211	4500	female	2007
		46.7		195	3300	female	2007
				190	0	female	2008
			18.4	200	0	female	2008
Chinstrap	Biscoe	5	19	200	0	male	2008
...	...	...	...	...	...	...	...









from www.kaggle.com

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... the example through DIKW and CRISP-DM

penguins as a first “controlled” dataset

DIKW	***in the penguins dataset***
Data	`bill_length, flipper_length, island, sex, species, ...`
Information	cleaned and typed table; numerical summaries by species
Knowledge	a classifier that predicts species from observed measurements
Wisdom	is the model is useful enough for a given educational or analytical purpose?

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... limits and cautions

- DIKW is a conceptual **simplification**, not a strict (scientific) law
- CRISP-DM is **helpful**, but real work is often more iterative and less linear
- ... a process formulation is useful only if it supports better analytical decisions
- so, CRISP-DM will be treated as useful reference; not a rigid “dogma”!

- MLDM sees the process as a search for the **best** defensible **data-to-model** path
- ... a perspective where alternative paths, leakage, validation and deploy are central

A final “one sentence” message



DIKW clarifies the **nature of transformation**

CRISP-DM provides a classical **process view of how that transformation is organised**

together they prepare **MLDM** to be **understood as a search for the best defensible data-to-model path**

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